

flowchart

task 1



go step by step and identify boolean algebra properties in the given equation



simplify expression when found

task 2



make a truth table
to get input/output relationships



write expression



use boolean algebra properties
to simplify expression.

I did not experience much difficulty in this activity as these are topics that I've learned before. Though, I will say that I much prefer using K-maps instead of simplifying by hand.

①

$$\begin{aligned}
 F &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C \\
 &= \bar{A}\bar{B}(\bar{C} + C) + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C \\
 &= \bar{A}(\bar{B} + B\bar{C}) + A\bar{B}\bar{C} + A\bar{B}C \\
 &= \bar{A}(\bar{B} + C) + A\bar{B}\bar{C} + A\bar{B}C \\
 &= \bar{A}\bar{B} + \bar{A}C + A\bar{B}\bar{C} + A\bar{B}C \\
 &= \bar{A}\bar{B} + C(\bar{A} + A\bar{B}) + A\bar{B}C \\
 &= \bar{A}\bar{B} + C(\bar{A} + \bar{B}) + A\bar{B}C \\
 &= \bar{A}\bar{B} + \bar{A}C + \bar{B}C + A\bar{B}C
 \end{aligned}$$

②

m	t	i
0	0	0
0	1	1
1	0	1
1	1	1

$$\begin{aligned}
 F &= \bar{m}t + m\bar{t} + mt \\
 &= \bar{m}t + m(\bar{t} + t) \\
 &= \bar{m}t + m \\
 &= t + m \quad \text{OR gate}
 \end{aligned}$$